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Continental Shelf Research 27 (2007) 1477–1500

CONTINENTAL SHELF RESEARCH

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Temporal and spatial variability of radium in the coastal ocean and its impact on computation of nearshore cross-shelf mixing rates

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Received 12 March 2006; received in revised form 13 December 2006; accepted 8 January 2007
Available online 18 January 2007

Abstract

Constraining the exchange of water from the shoreline to the mid-shelf is necessary for the development of accurate and predictive models of nearshore circulation. Ra isotopes, which emanate from sediments and have a variety of half-lives, may be useful in measuring cross-shelf mixing rates. The distributions of Ra isotopes were measured in transects extending perpendicular from the shoreline at Sunset Beach and Huntington Beach, CA. The average inventory at Sunset Beach was four times greater than at Huntington Beach. Building on previous research on Ra inputs and circulation in San Pedro Bay, a two-dimensional model for surface water Ra was developed to identify the importance of onshore flow and cross-shelf mixing near Huntington Beach. For the mean summertime conditions, the eddy diffusivity (K_h) was $1.4 \pm 0.4\,m^2\,s^{-1}$, with 8% of the water from Sunset Beach moving down the coast. The remaining water must be low-Ra water that has moved onshore. At time scales greater than a week, the short-lived Ra inventory at Huntington Beach varied by 50%, which reflects changes in the fractions of water moving down-coast and/or in the longshore advection rate. The shoreline Ra concentration varied on time scales of hours, which may be generated by tidal changes in the Ra input at the shoreline and short-period fluctuations in the mixing rate. The low K_h observed in this study in comparison to higher values measured further offshore is evidence that K_h increases with distance offshore. When scale-dependent mixing beyond 455 m offshore is incorporated into the model, the results are consistent with the observed data for ^{223}Ra, ^{224}Ra, and ^{228}Ra. Using the model, the ^{228}Ra input flux to the summertime mixed layer was between 3.4×10^6 and 4.0×10^6 atoms s^{-1} (m shoreline)$^{-1}$.
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Keywords: Radium; ^{224}Ra; ^{223}Ra; ^{228}Ra; Coastal ocean; Eddy diffusivity; USA; CA; San Pedro Bay; Huntington Beach

1. Introduction

Beach and coastal ocean management require accurate, predictive models that track the fate of contaminants and nutrients from the shoreline to the open ocean. However, most ocean circulation models stop at the 10 m isobath, ignoring inner shelf and surf zone processes where rivers, tidal marshes, storm drains, and shallow groundwater aquifers discharge. Nearshore circulation models are needed to accurately connect these sources with regional circulation models. To do this, the nearshore transport of water and solutes must be quantified.

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doi:10.1016/j.csr.2007.01.003

Because of slow mean cross-shelf flow rates, eddy diffusion plays an important role in transporting contaminants and nutrients from the shoreline to the middle shelf. Quantifying eddy diffusion evaluates the truly turbulent part of a mean flow. One method to calculate diffusivity in nearshore surface waters is based on the distribution of short-lived radium isotopes, ^{223}Ra and ^{224}Ra (Moore, 2000; Torgersen et al., 1996). To do this effectively, a mass balance for Ra in the coastal ocean must be developed to account for the various Ra sources that may affect the Ra distribution.

Radium has four naturally occurring isotopes, with a broad range of half-lives: ^{223}Ra (11.4 days), ^{224}Ra (3.66 days), ^{226}Ra (1600 years), and ^{228}Ra (5.75 years). Each is produced by a thorium isotope parent that is either bound within the structure of minerals or adsorbed to sediment surfaces. As a result, margin sediments are the primary source of Ra to the ocean. The distribution of each isotope in the coastal ocean is a function of its inputs, the rate of offshore transport, and the rate of radioactive decay. The short-lived isotopes, ^{223}Ra and ^{224}Ra, have half-lives on the order of time constants for coastal ocean processes, and therefore should be useful in calculating nearshore mixing rates. The longer lived isotopes provide additional constraints on the hydrologic budgets.

Much of the coastal ocean application of short-lived Ra isotopes has been conducted in the south Atlantic bight, where the first coastal ocean measurements of ^{224}Ra were made (Levy and Moore, 1985). Continuing research in the region, Moore (2000) developed a model to compute the eddy diffusivity from the short-lived Ra distribution. Once Ra is added to nearshore waters, cross-shelf mixing will transport it away from its source, and the distance traveled before decay will depend on the isotope's half-life. Assuming Ra behaves conservatively in the mixed layer, this can be expressed with a differential equation:

$$\frac{\partial C}{\partial t} = K_h \frac{\partial^2 C}{\partial x^2} - \lambda C, \tag{1}$$

where C is the isotope concentration, t the time, K_h the the eddy diffusivity, x the distance offshore, and λ the isotope's decay constant. Eq. (1) simplifies the Ra dynamics in the mixed layer by assuming that Ra inputs, longshore gradients of Ra, cross-shelf advection, and vertical mixing are all negligible. To solve, we assume that a constant input flux at the landward boundary generates a constant concentration, C_o, and no excess Ra very far offshore. If K_h is constant and the system is in steady state,

$$C(x) = C_o \, e^{-x\sqrt{\lambda/K_h}}. \tag{2}$$

Fitting this equation to the short-lived Ra data, the cross-shelf mixing rate K_h can be determined. This method was successfully applied to fit short-lived Ra data collected beyond 10 km offshore in the mid-Atlantic bight (Moore, 2000).

For the nearshore mixed layer, several of the assumptions inherent in this model are violated. First, Ra inputs are not localized at the shoreline, and instead occur across the seafloor in contact with the mixed layer (Colbert and Hammond, 2007). Second, longshore variability of Ra sources, such as the presence of estuaries, and sinks, such as onshore flows, may impact the Ra budget. Third, eddy diffusion in the open ocean is generally not constant, but instead is scale dependent (Okubo, 1971). As a result, this model generates inconsistent results for ^{223}Ra and ^{224}Ra (e.g. Boehm et al., 2006).

Addressing these three assumptions is the focus of this paper. For this study, the nearshore Ra distribution in surface water was measured several times during the summer at the north and south end of San Pedro Bay, CA. These measurements allow the temporal and spatial variability of the shoreline Ra concentration and the surface water inventory to be assessed. Next, Ra inputs from estuaries are constrained. Using this and previous estimates of seafloor and shoreline Ra inputs measured at Huntington Beach, CA (Colbert and Hammond, 2007), the relative importance of each short-lived Ra source and its longshore variability is examined. With known inputs and surface water distribution, a two-dimensional (2D) model for San Pedro Bay surface waters is developed to estimate the cross-shelf mixing rate and importance of onshore flow. Then, by varying the model parameters, sources of variability of the offshore Ra isotopes distribution are assessed. Finally, the model is applied to assess the input flux of ^{228}Ra. This model demonstrates some of the complexities inherent in using short-lived Ra isotopes as a tracer of coastal ocean mixing.

2. Study site

San Pedro Bay is a relatively broad (width $= 20$ km) continental shelf located near the middle of the Southern California Bight (Fig. 1), bounded by Palos Verdes Peninsula and Newport Canyon. At the north end of the bay is the Los Angeles-Long Beach Harbor, which is protected by a 14 km breakwall that extends east from Palos Verdes. Exchange between San Pedro Bay and the harbor complex occurs through three openings in the breakwall. Diurnal winds drive eastward net currents in surface water of the Outer Harbor, but at depth a westward flow is apparent (Mc Gehee et al., 1989).

Three rivers discharge into the Bay: the Los Angeles, San Gabriel, and Santa Ana Rivers. Most of the river water is diverted upstream for groundwater recharge. During the warm, dry summers, the rivers have a minor flow below these diversions composed of dry-weather runoff, treated domestic sewage, and industrial discharges. River discharge during the wet, mild winters is the primary source of sediment to San Pedro Bay (Emery, 1960).

Most of the water column density structure is defined by temperature. During the summer, a warm (18–21 °C), 5–15 m thick mixed layer develops. Below this are deep shelf waters that are 13–14 °C. Reduced winds during the summer and high solar radiation help maintain this profile. However the mixed-layer development often occurs

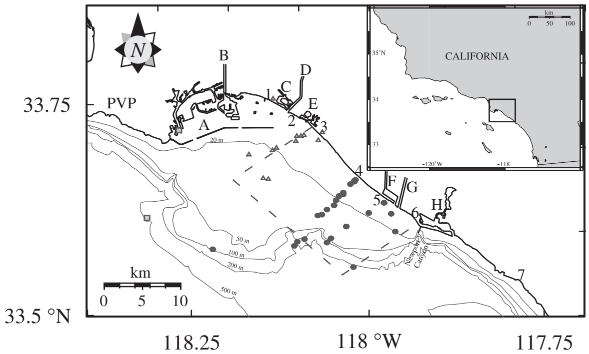

Fig 1. Map of San Pedro Bay, CA with inset map of Southern California. Santa Catalina Island is at the southwest corner of box in inset. Points and triangles represent sample stations offshore from Huntington Beach and Sunset Beach, respectively. Dotted line represents the boundaries of mixing model. Squares are buoy stations: NOAA Outer Los Angeles Harbor Station 9410660 and Scripps San Pedro Buoy 092. Bodies of water: A = Ports of Los Angeles and Long Beach, B = Los Angeles River, C = Alamitos Bay, D = San Gabriel River, E = Anaheim Bay, F = Talbert Marsh, G = Santa Ana River, H = Newport Bay. Cities and sample locations: 1 = Long Beach, 2 = Seal Beach, 3 = Sunset Beach, 4 = Huntington Beach, 5 = Huntington State Beach, 6 = Newport Beach, 7 = Laguna Beach, PVP = Palos Verdes Peninsula.

more rapidly than can be explained by solar radiation, implying the importance of the advection of warm water masses into the region (Noble et al., 2003).

Tides in Southern California are mixed, with two high tides and two low tides of unequal height. The mean tide is approximately 1.1 m; the spring tides exceed 2 m. Across much of the shelf, the major tidal currents flow parallel to the shoreline. Tidal currents propagate to the north and have strong oscillatory motions that result in little net movement over a tidal cycle (Karl et al., 1980; Noble et al., 2003).

Total flow along the shelf and through the San Pedro Channel at the north end of San Pedro Bay is dominated by the northward-flowing Southern California Countercurrent (Hickey, 1993). During the summer, however, surface slope currents generally flow towards the south, above a northward-flowing undercurrent (Hickey, 1993; Noble et al., 2003). On the shelf, currents follow the direction of local isobaths and are geostrophic in character (Hendricks, 1994; Hickey, 1993; Karl et al., 1980; Noble et al., 2003). During summer, shelf surface currents (< 15 m) typically flow towards the south (Hickey, 1993; Noble et al., 2003; Winant and Bratkovich, 1981). A critical constraint on circulation in the area is provided by an extended current meter deployment near Huntington Beach (Noble et al., 2003). Mean mixed-layer flow was southeast, parallel to the shore, averaging 5 cm s^{-1}.

3. Methods

3.1. Water sampling

Nine transects were made extending offshore from Huntington Beach, and four transects were made at Sunset Beach from the R/V Harmony. Typical transects contained three stations plus one at the shoreline. Salinity and temperature profiles were collected at each station using a hand-held CTD (Yellow Springs Instr. Model 85). The mixed-layer thickness was defined as the depth from the surface to where the temperature decreased by 1 °C.

Ra samples were generally collected at three depths per station, by deploying a hose to the desired sample depth and pumping the water to the surface. At the shoreline, samples were collected with buckets in knee-deep water every 2 h for 8 h—a total of five samples each day. About 100 L of seawater was pumped through a 5 μm filter to remove particulates, a flow totalizer to measure the water volume, and then through a cartridge containing acrylic fibers impregnated with manganese dioxide. Ra is extracted from sea water onto these Mn-fibers with high efficiency (Moore, 1976).

3.2. Radium analysis

Within 4 days of collection, ^{223}Ra and ^{224}Ra adsorbed to the acrylic Mn-fibers were analyzed using a flow-through system with a scintillation detector linked to a time-delayed coincidence counter (RaDeCC) that can distinguish the decays of the Ra daughters, ^{219}Rn and ^{220}Rn, which rapidly grow into equilibrium with their parents (Moore and Arnold, 1996). The measured activities were corrected for decay and ^{224}Ra ingrowth from ^{228}Th that also adsorbed to the fibers. Some fibers were stored for 6 months or more, allowing ^{224}Ra and ^{228}Th ingrowth from ^{228}Ra. These were then recounted and used to determine ^{228}Ra activity.

Two methods were used to standardize RaDeCC. First, a standard was made by dissolving known masses of EPA monazite and pitchblende standards and passing the solution through a Mn-fiber filter to sorb Ra and its Th parent. Second, a few samples were processed and measured by gamma-ray spectrometry (Elsinger et al., 1982; Moore, 1984). After measuring the sample on the RaDeCC, fibers were soaked in a 90 °C 1 N NH_2OH–HCl solution to dissolve the Mn-oxide and release the Ra from the fibers, with 2 g of $BaCl_2$ added. Once the fibers turned white, the solution was cooled and filtered to remove any fiber. To precipitate $BaSO_4$, which efficiently incorporates Ra into its structure, 25 mL of a 20% H_2SO_4 solution was added. The precipitate was filtered from the solution, dried, and transferred to a test tube. The gamma-ray activity of ^{224}Ra and its daughter, ^{212}Pb, as well as ^{228}Ac, ^{214}Pb, and ^{214}Bi, were measured using a HP-Ge well-type gamma spectrometer. Absolute ^{223}Ra, ^{224}Ra, and ^{228}Ra activities were computed based on EPA diluted monazite and diluted pitchblende standards. Results of the two methodologies agreed within analytical uncertainties.

Samples for ^{226}Ra analysis were collected in 20 L glass carboys. ^{226}Ra was measured based on ^{222}Rn ingrowth (Mathieu et al., 1988). The efficiencies of each extraction system and counting cell have been computed based on GEOSECS, EPA, and internal standards.

4. Results

4.1. Shoreline

Surf zone short-lived Ra concentrations are related to the tides, with higher concentrations during low tide (Fig. 2). Daily isotope concentration fluctuations were usually about 25%, indicating the complex interaction between Ra inputs from water circulation through beach and offshore removal. The daily ^{224}Ra/^{223}Ra ratio remained relatively constant, indicating that the two isotopes probably have a coherent source.

On 4 sampling days, the water temperature at the shoreline was also measured (Fig. 3). In general, the temperature increased throughout the day, and began to cool by the time the last sample was collected, after 16:00 local time. There was no clear correlation with the tides or Ra concentration. This is consistent with a nearshore increase in temperature observed at buoys offshore from Huntington Beach and attributed to diurnal winds (Noble et al., 2003). In general, the surf zone temperature was within the temperature range of the mixed layer. Upwelling from below the mixed layer does not appear to be an important source of water in the surf zone.

4.2. Surface water distribution

Short-lived Ra in the surface mixed layer was measured at Huntington Beach (Table 1A) and at Sunset Beach (Table 1B). Samples collected beyond 7 km offshore were assumed to represent the supported activity for ^{223}Ra and ^{224}Ra and indicate that virtually all the nearshore activities of these two isotopes are unsupported. The only sampling day during wintertime conditions was on April 3, 2002. Because of the importance of other processes during winter, such as vertical mixing, the following discussion will be based solely on the summertime data, collected between June and September. In Fig. 4, four representative profiles from Huntington Beach are presented. Sampling was done to cover a range of tide and wave conditions, as well as differences in local and regional circulation that may influence nearshore circulation. Therefore, these profiles should capture most of the variability in the system. In general, high concentrations were observed at the shoreline that decreased rapidly within the first few km offshore. This is far different from the mid-Atlantic bight, where ^{224}Ra

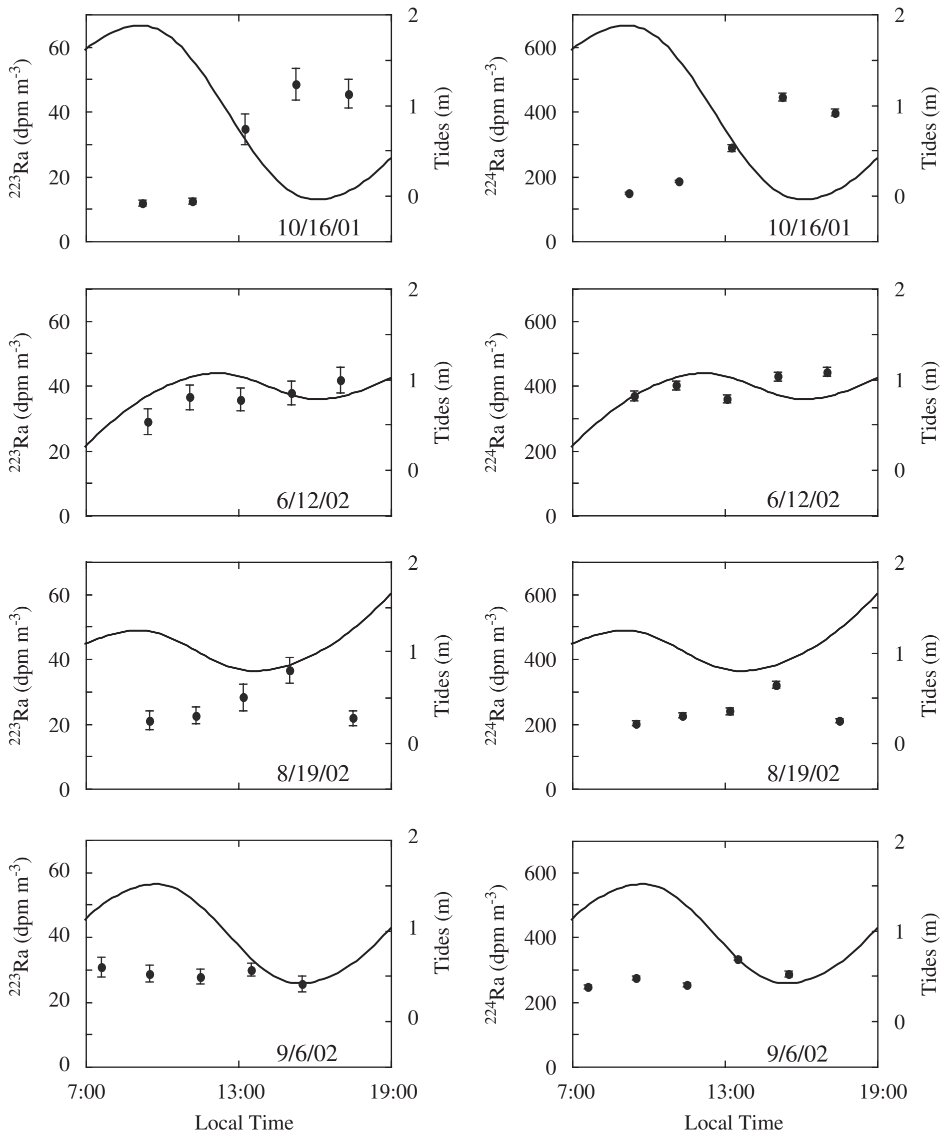

Fig. 2. Representative shoreline concentrations of ^{223}Ra (left column) and ^{224}Ra (right column) at Huntington Beach on 4 days. Tidal height (solid line) taken from tide tables.

concentrations greater than 100 dpm m^{-3} are measured 10 km offshore, and background concentrations are only reached beyond 50 km offshore (Levy and Moore, 1985; Moore, 2000).

^{228}Ra was measured on a subset of samples (Table 1). There was no significant difference in surface water ^{228}Ra activity between Sunset Beach and Huntington Beach, so all samples collected in San Pedro Bay are presented together (Fig. 5). Similar to the short-lived isotopes, the highest ^{228}Ra activities were observed at the shoreline and decreased with distance offshore. Off Southern California, ^{228}Ra concentrations continue to decrease for tens of km offshore (Knauss et al., 1978).

4.3. Surface water inventory

The short-lived Ra isotope inventory I (dpm (m shoreline)$^{-1}$) in surface waters was calculated by integrating the activity offshore, assuming the mixed-layer activity (λC) is independent of depth and taking into account the increasing depth of

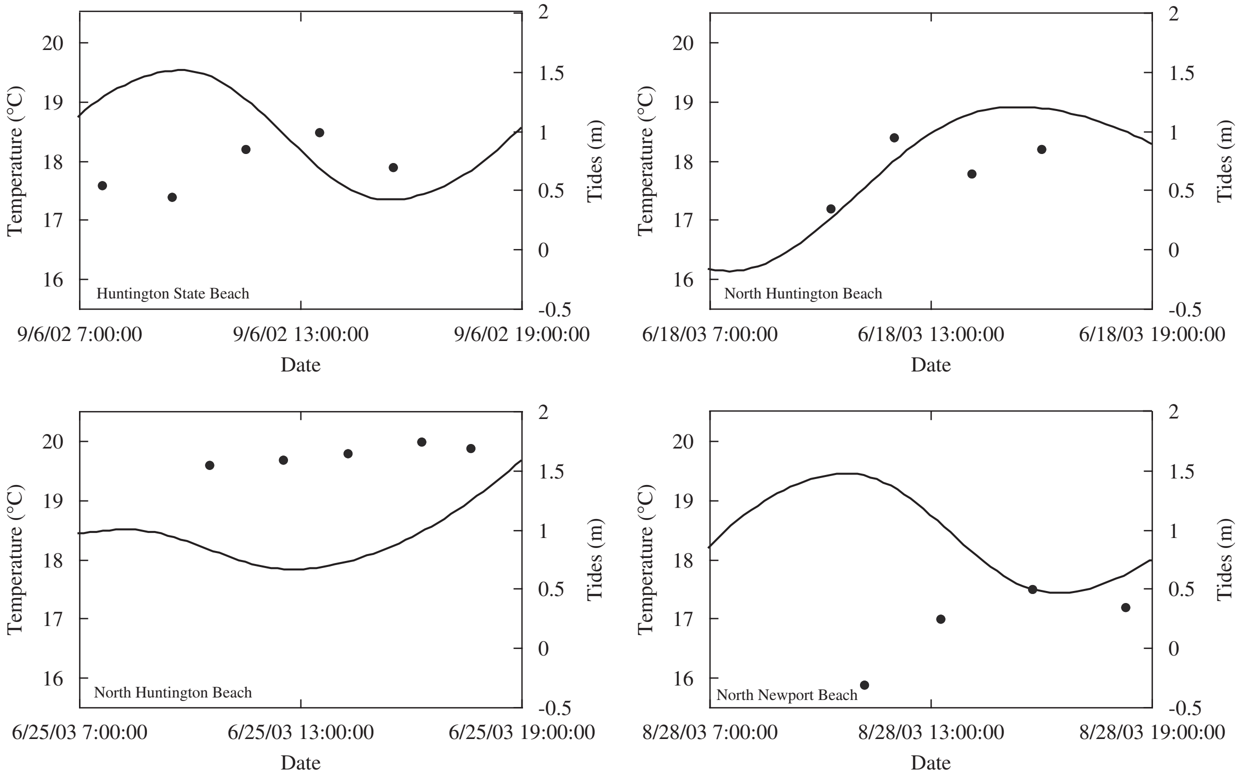

Fig. 3. Shoreline temperature as a function of time. Tidal height (solid line) taken from tide tables.

water with distance offshore:

$$I = \int_0^\infty \int_0^{H_m} \lambda C \, dz \, dx, \qquad (3)$$

where H_m, the mixed-layer thickness, is a function of the distance offshore x, and constrained by water depth, z. Eq. (2) was used to interpolate the concentrations between sampling stations. The mixing rates derived from this fit will be further discussed in Section 6. Using an average H_m of 11.8 ± 4.0 m, the mean short-lived Ra inventory was computed for the Huntington Beach and Sunset Beach transects (Table 2). Sampling at Sunset Beach was usually conducted within 2 days of sampling at Huntington Beach. The inventory at Sunset Beach was 3–4 times greater than at Huntington Beach. These average profiles are further used to develop the 2D mixing model presented below.

Fitting Eq. (2) to the daily profiles, the temporal variability of the short-lived Ra surface water inventory was assessed at Huntington Beach (Table 2). Over periods of days to a week, the inventories remained relatively stable. Over longer periods, significant Ra inventory changes were observed. These time scales are similar to fluctuations in flow measured with current meters on time scales of 5–10 days (Noble et al., 2003).

4.4. Tidal exchange at estuaries

During summer, river freshwater inputs to San Pedro Bay are small, and exchange between rivers and the coastal ocean is dominated by tidal flushing. To compute the Ra flux to the nearshore from estuaries, the net exchange of Ra at the mouth of each tidal inlet was measured. This method accurately estimates the flux of short-lived Ra isotopes out of estuaries (Hancock et al., 2000).

At the south end of Huntington Beach is the Santa Ana River and the Talbert Marsh. In these systems, the volume of water exchanged on a tidal cycle has been computed, and the storage was found to scale with the same proportion to the surface area (Grant et al., 2003). The storage to surface area ratio is equivalent to the mean tidal height of about 1 m. To compute the tidal exchange volume, the surface area of Alamitos Bay, Anaheim Bay, Newport Bay, and the San Gabriel River was measured from USGS

Table 1
Short-lived Ra isotope concentrations from (A) shoreline and mixed layer off Huntington Beach, and (B) Sunset Beach

Collect date, time	Mixed layer thick. (m)	Sample depth (m)	Distance offshore (km)	Concentration ($dpm\,m^{-3}$)		
				^{223}Ra	^{224}Ra	^{228}Ra
(A) Huntington Beach shoreline and offshore transects						
10/19/2000, 16:05		0.2	0.00			89 ± 16
10/19/2000, 17:25		0.2	0.00			111 ± 24
10/19/2000, 18:45		0.2	0.00			197 ± 36
10/19/2000, 8:45		0.5	0.78			84 ± 8
10/19/2000, 10:00		0.5	0.71			108 ± 10
10/19/2000, 10:45		0.5	0.70			102 ± 10
12/1/2000, 14:10		0.2	0.00			169 ± 20
12/1/2000, 15:00		0.2	0.00			153 ± 18
12/1/2000, 15:55		0.2	0.00			279 ± 29
12/1/2000, 9:20		0.5	0.74			117 ± 15
12/1/2000, 10:20		0.5	0.65			109 ± 13
12/1/2000, 11:05		0.5	2.43			78 ± 11
9/13/2001, 16:53	8.5	1.5	1.35	8.3 ± 2.0	83.5 ± 5.1	52 ± 7
9/13/2001, 14:50	>10	1.5	2.97	1.3 ± 0.5	13.9 ± 1.4	32 ± 8
9/13/2001, 12:49	>12.1	1.5	9.86	0.1 ± 0.1	0.6 ± 0.1	6 ± 5
9/13/2001, 9:16	>12.1	1.5	19.98	0.1 ± 0.1	0.4 ± 0.1	1 ± 7
10/16/2001, 9:12		0	0	12.0 ± 0.9	150.1 ± 3.3	
10/16/2001, 11:10		0	0	12.5 ± 0.8	187.2 ± 3.6	
10/16/2001, 13:15		0	0	34.7 ± 4.7	290.2 ± 10.6	
10/16/2001, 15:15		0	0	48.6 ± 4.9	446.9 ± 11.5	
10/16/2001, 17:20		0	0	45.6 ± 4.4	399.0 ± 10.2	
4/3/2002, 9:45		1.5	1.67	22.9 ± 1.6	258.7 ± 5.4	81 ± 14
4/3/2002, 12:15		1.5	4.41	11.1 ± 1.4	86.5 ± 3.4	38 ± 10
4/3/2002, 14:15		1.5	7.48	0.9 ± 0.2	3.8 ± 0.3	10 ± 7
4/3/2002, 15:45		1.5	11.89	2.3 ± 0.3	9.6 ± 0.6	8 ± 7
6/7/2002, 9:00		0	0	32.3 ± 6.1	264.6 ± 15.4	153 ± 4
6/7/2002, 10:45		0	0	48.7 ± 6.5	423.7 ± 17.4	154 ± 4
6/7/2002, 12:40		0	0	50.7 ± 6.3	349.5 ± 14.5	140 ± 3
6/7/2002, 14:35		0	0	65.1 ± 9.4	635.8 ± 26.4	
6/7/2002, 16:35		0	0	25.5 ± 5.4	324.4 ± 17.1	100 ± 2
6/7/2002, 10:05	7.0	1.5	1.08	13.5 ± 1.0	132.2 ± 3.0	
6/7/2002, 12:30	>15	1.5	3.77	0.6 ± 0.2	5.4 ± 0.6	
6/7/2002, 14:20	20.0	1.5	7.09	0.2 ± 0.1	0.7 ± 0.1	
6/7/2002, 16:50	10.0	1.5	12.48	0.1 ± 0.1	0.4 ± 0.1	
6/12/2002, 9:25		0	0	29.1 ± 3.9	369.9 ± 14.1	69 ± 2
6/12/2002, 11:05		0	0	36.6 ± 3.9	403.4 ± 13.6	78 ± 2
6/12/2002, 13:05		0	0	35.8 ± 3.5	360.9 ± 10.9	64 ± 2
6/12/2002, 15:05		0	0	38.0 ± 3.7	430.2 ± 13.0	79 ± 2
6/12/2002, 17:00		0	0	41.8 ± 4.0	444.8 ± 13.7	82 ± 2
6/12/2002, 10:00	9.1	1.5	0.99	12.7 ± 1.1	147.5 ± 3.6	
6/12/2002, 12:00	>17	1.5	3.77	0.6 ± 0.1	5.0 ± 0.3	14 ± 7
6/12/2002, 14:20	22.0	1.5	6.12	0.2 ± 0.1	0.9 ± 0.1	
6/12/2002, 16:45	>15	1.5	10.97	0.1 ± 0.3	0.9 ± 0.2	
7/26/2002, 7:50		0	0	49.2 ± 3.8	553.8 ± 13.1	
7/26/2002, 9:50		0	0	33.6 ± 2.9	389.8 ± 10.8	
7/26/2002, 11:55		0	0	36.0 ± 3.0	361.0 ± 9.3	
7/26/2002, 14:05		0	0	65.7 ± 6.3	726.4 ± 21.2	

Table 1 (*continued*)

Collect date, time	Mixed layer thick. (m)	Sample depth (m)	Distance offshore (km)	Concentration ($dpm\,m^{-3}$)		
				^{223}Ra	^{224}Ra	^{228}Ra
7/26/2002, 15:55		0	0	30.7±2.4	394.0±8.7	
7/26/2002, 13:20	8.0	1.5	0.80	10.6±0.9	78.1±2.7	
7/26/2002, 14:10	5.0	1.5	2.90	11.9±0.9	80.2±2.6	
8/19/2002, 9:30		0	0	21.2±2.8	202.8±7.7	70±16
8/19/2002, 11:20		0	0	22.8±2.7	227.8±7.6	77±16
8/19/2002, 13:10		0	0	28.4±4.1	241.0±10.5	80±17
8/19/2002, 15:00		0	0	36.6±4.0	321.4±10.6	70±15
8/19/2002, 17:30		0	0	22.0±2.3	210.9±6.5	63±16
8/19/2002, 14:30	10.0	1.5	0.68	12.6±1.7	94.2±3.9	58±11
8/19/2002, 13:20	18.0	1.5	2.96	0.2±0.1	2.4±0.3	7±6
8/19/2002, 11:40	14.0	1.5	6.09	0.1±0.1	1.0±0.3	14±7
8/19/2002, 9:35	10.0	1.5	9.31	0.2±0.1	0.8±0.1	20±8
8/23/2002, 8:00		0	0	26.5±2.2	303.3±7.0	74±20
8/23/2002, 11:50		0	0	11.8±1.4	183.4±5.3	84±22
8/23/2002, 14:00		0	0	23.6±1.9	264.8±6.5	90±23
8/23/2002, 15:55		0	0	37.1±3.2	380.0±9.2	122±25
8/23/2002, 17:50		0	0	22.7±1.7	332.5±6.9	110±23
8/23/2002, 14:10	10.0	1.5	0.67	5.8±0.8	51.9±2.1	48±11
8/23/2002, 13:00	11.0	1.5	2.96	3.3±0.3	19.1±0.6	39±10
8/23/2002, 11:20	13.0	1.5	6.09	0.2±0.1	1.3±0.2	9±6
8/23/2002, 9:45	11.0	1.5	8.58	0.3±0.1	0.8±0.1	0±0
9/6/2002, 7:35		0	0	30.9±2.9	247.3±6.4	
9/6/2002, 9:30		0	0	28.9±2.5	273.9±6.0	
9/6/2002, 11:30		0	0	27.9±2.2	253.6±5.4	
9/6/2002, 13:30		0	0	30.0±2.0	334.6±5.6	
9/6/2002, 15:30		0	0	25.6±2.5	287.3±7.9	
6/18/2003, 12:00		0	0	19.6±3.8	168.4±9.4	61±21
6/18/2003, 14:05		0	0	21.0±2.7	270.4±8.4	
6/18/2003, 10:20	>10	1.5	0.94	9.2±0.5	64.2±1.1	53±10
6/18/2003, 12:30	14.0	1.5	3.28	1.3±0.1	3.9±0.2	
6/18/2003, 14:35	6.0	1.5	5.15	0.7±0.1	2.0±0.1	25±6
6/25/2003, 10:30		0	0	26.0±3.0	243.3±7.5	108±23
6/25/2003, 12:30		0	0	28.6±2.7	373.3±8.1	113±24
6/25/2003, 14:15		0	0	25.3±2.6	255.9±6.8	
6/25/2003, 16:15		0	0	31.1±2.7	370.6±7.9	
6/25/2003, 17:35		0	0	24.7±3.6	242.4±9.7	
6/25/2003, 9:30	8.0	1.5	1.10	10.5±0.8	77.3±1.7	60±13
6/25/2003, 11:10	9.0	1.5	3.29	6.9±0.6	33.6±1.0	
6/25/2003, 13:10	10.5	1.5	5.16	0.8±0.1	4.5±0.2	
8/28/2003, 11:10		0	0	41.5±4.1	416.6±11.8	
8/28/2003, 15:45		0	0	18.4±1.5	237.3±4.5	
8/28/2003, 18:15		0	0	34.1±2.3	392.8±7.3	
(B) Other shoreline and offshore transects						
Sunset Beach						
9/11/2001, 11:10	7	1.5	2.20	0.5±0.6	6.0±1.5	35±2
9/11/2001, 13:53	7	1.5	5.50	11.6±2.1	31.7±2.6	35±2
9/11/2001, 17:47	6	1.5	9.10	0.8±0.2	2.2±0.3	26±2
9/13/2001, 9:16	>12.1	1.5	14.00	0.1±0.1	0.4±0.1	

Table 1 (*continued*)

Collect date, time	Mixed layer thick. (m)	Sample depth (m)	Distance offshore (km)	Concentration ($dpm\,m^{-3}$)		
				^{223}Ra	^{224}Ra	^{228}Ra
Sunset Beach						
9/20/2001, 12:55	9	1.5	1.70	18.4±2.7	114.1±5.0	61±1
9/20/2001, 10:51	11	1.5	4.90	10.9±2.7	42.6±4.3	45±2
9/20/2001, 8:44	>8.6	1.5	9.10	2.5±0.3	11.6±0.5	18±1
Belmont Shore						
9/25/2001, 8:35		0	0	23.3±3.7	256.0±10.1	65±3
9/25/2001, 10:40		0	0	52.0±8.1	345.0±16.3	90±3
9/25/2001, 12:35		0	0	48.7±4.2	327.1±8.3	89±2
9/25/2001, 14:35		0	0	61.9±7.4	277.1±11.7	86±2
9/25/2001, 16:35		0	0	32.4±4.1	278.1±9.2	
9/25/2001, 18:35		0	0	36.5±4.5	290.7±9.8	91±4
Sunset Beach						
6/5/2002, 8:40		0	0	55.0±5.7	457.2±12.9	
6/5/2002, 10:45		0	0	58.7±5.0	474.4±11.4	
6/5/2002, 12:40		0	0	76.0±10.5	485.0±20.8	
6/5/2002, 14:35		0	0	95.5±10.7	671.3±22.6	
Sunset Beach						
6/5/2002, 15:55	12	1.5	1.9	39.2±6.0	244.6±11.8	
6/5/2002, 13:45	>12	1.5	2.89	23.1±1.0	52.1±1.2	
6/5/2002, 10:30	12	1.5	7.28	25.8±1.4	259.2±3.7	
Sunset Beach						
7/26/2002, 8:00		0	0	46.8±3.4	805.4±19.8	131±5
7/26/2002, 9:50		0	0	49.5±3.8	755.1±21.3	122±5
7/26/2002, 11:50		0	0	44.9±2.9	632.0±15.1	96±4
7/26/2002, 13:50		0	0	64.8±5.0	804.9±23.0	141±8
7/26/2002, 15:50		0	0	70.5±3.3	637.4±13.4	109±6
7/26/2002, 15:30	4	1.5	0.72	23.3±2.4	219.3±7.0	67±2
Newport Beach						
7/26/2002, 8:30		0	0	20.5±2.0	219.1±6.7	
7/26/2002, 10:20		0	0	19.4±1.1	223.1±5.0	
7/26/2002, 12:15		0	0	17.5±1.9	185.0±5.8	
7/26/2002, 14:10		0	0	21.7±2.0	185.0±5.9	
7/26/2002, 15:55		0	0	23.9±2.5	222.2±8.3	
7/26/2002, 11:20		1.5	0.4	10.5±1.0	90.0±2.9	

Values are organized based on collection date. Distance offshore is measured from the mean low water mark. Uncertainty is 1 standard deviation based on the counting statistics.

topographic maps, and the surface area to volume ratio was applied (Table 3). For Anaheim Bay, this estimate represents an upper limit because tidal flow in parts of the bay is restricted by culverts and tide gates (CalDeptF&G, 1976).

Water flowing into the estuary has an initial activity, which can be estimated based on the average nearshore distribution of short-lived Ra isotopes. For estuaries that discharge into the surf zone, the average local shoreline activity was used (Table 3). For estuaries with discharge channeled through offshore jetties up to 0.9 km offshore, the activity at the mouth of the jetty was estimated from Fig. 6. Samples were collected at the mouth of each estuary during the ebb tide and averaged (Table 3). The tidal exchange flux from each estuary was computed as the product of the water exchange rate and the difference in activity between ebb and flood waters. The greatest source of uncertainty is the Ra activity difference between the inflow and outflow, which was greatest for estuaries that discharge at the shoreline.

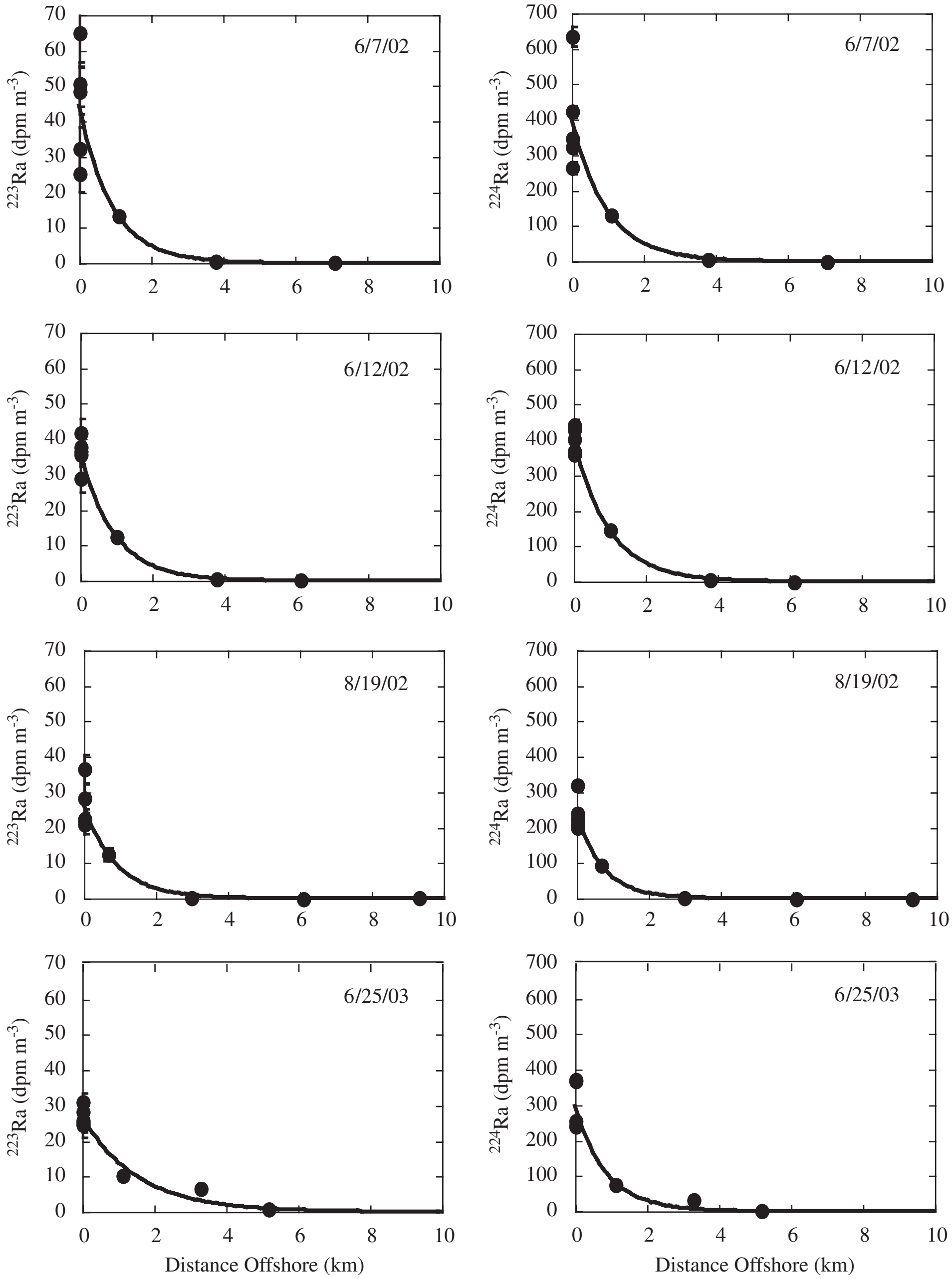

Fig. 4. Nearshore distribution of ^{223}Ra (left column) and ^{224}Ra (right column) on 4 representative sampling days at Huntington Beach. When not shown, error bars are smaller than point. Exponential fit (Eq. (2)) through data is shown. Model fit parameters are presented in Table 2.

4.5. Ra sources

The Ra inputs from the shoreline and seafloor at Huntington Beach were measured, and the long-shore variability of Ra inputs from these sources was assessed (Colbert and Hammond, 2007). This study found that the short-lived Ra emanation rate was roughly proportional to the total sediment ^{238}U and ^{232}Th activity. In San Pedro Bay, from just southeast of Sunset Beach to Newport Canyon, the

^{238}U and ^{232}Th production rates, and therefore short-lived Ra, were relatively constant. For San Pedro Bay, the primary source of ^{223}Ra and ^{224}Ra is the seafloor, while the shoreline and tidal exchange with the Talbert Marsh and Santa Ana River are minor sources (Table 4). At the north end of San Pedro Bay, shoreline and seafloor sources of Ra are augmented by exchange with the Los Angeles/Long Beach Harbor and inputs from the Los Angeles and San Gabriel Rivers, Alamitos Bay, and Anaheim Bay. These latter sources are difficult to evaluate but

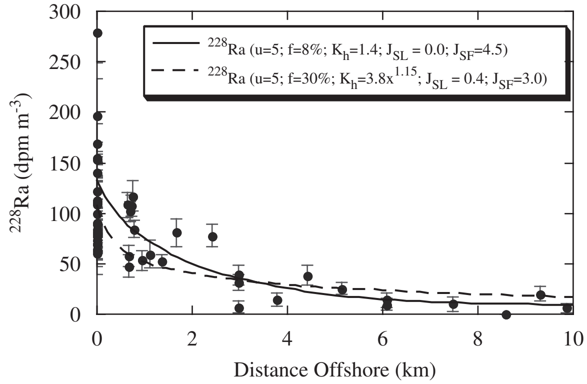

Fig. 5. Composite summertime ^{228}Ra at Huntington Beach with model best-fit results for constant-K_h (solid line) and scale-dependent-K_h (dashed line). The fractional inventory of water from Sunset Beach and the diffusivity were set based on the best-fit model results for ^{223}Ra and ^{224}Ra for longshore advection rate of $5\,cm\,s^{-1}$. The seafloor (J_{SF}) and shoreline (J_{SL}) flux (10^6 atoms s^{-1} (m shoreline)$^{-1}$) were adjusted to find the best fit to the data.

sufficient to produce inventories for both isotopes in surface water at Sunset Beach that are 3–4 times greater than observed at Huntington Beach.

South of Newport Canyon, the ^{238}U and ^{232}Th total sediment activities decreased by 50%. In addition, the seafloor slope increases, reducing the seafloor–mixed layer contact area. As a result, Ra inputs from the shoreline and seafloor are significantly reduced (Table 5). South of San Pedro Bay, the primary short-lived Ra source is Newport Bay. Both in San Pedro Bay and south of Newport Canyon, the relative importance of each source is approximately the same for both isotopes.

It is evident from mass balances for the short-lived isotopes at Huntington Beach that onshore flow to the north dilutes Ra-rich water found at Sunset Beach (Colbert and Hammond, 2007). This flow is likely to vary temporally and cause variations in the longshore transport through the study area. It is unlikely that the inventory fluctuations (Table 2) were generated by changes in the Ra input flux. Based on the depth and rate of pore water exchange in these sediments, pore water exchange is controlled by physical processes, such as tidal height, wave height, or mixed-layer thickness (Colbert, 2004). No correlation was found between the inventory and environmental conditions that may affect the benthic Ra fluxes (Colbert, 2004). In addition, changes in the input should be independent of the mixing rate K_h. Instead, the inventory and the one-dimensional (1D) estimate of K_h are proportional (Table 2). These factors can only be

Table 2
Model-derived values for average shoreline concentration and scale distance, along with the computed inventory

Collect date	^{223}Ra Co ($dpm\,m^{-3}$)	Scale dist. (km)	K_h ($m^2\,s^{-1}$)	^{223}Ra inventory ($10^6\,dpm\,m^{-1}$)	^{224}Ra Co ($dpm\,m^{-3}$)	Scale dist. (km)	K_h ($m^2\,s^{-1}$)	^{224}Ra inventory ($10^6\,dpm\,m^{-1}$)
9/13/2001	38.6±2.4	0.88±0.03	0.5±0.1	0.21±0.02	371±12	0.90±0.02	1.8±0.1	2.16±0.08
4/3/2002	39.8±7.5	3.08±0.71	6.7±2.2	1.18±0.35	540±67	2.28±0.31	11.6±2.2	11.18±2.05
6/7/2002	44.5±5.3	0.91±0.67	0.6±0.6	0.26±0.19	400±49	0.97±0.71	2.1±2.2	2.59±1.93
6/12/2002	36.3±1.6	0.94±0.24	0.6±0.2	0.22±0.06	402±12	0.98±0.18	2.1±0.6	2.64±0.50
7/26/2002	43.0±6.2	0.63±0.57	0.3±0.4	0.14±0.13	485±64	0.45±0.44	0.4±0.6	0.98±0.98
8/19/2002	26.2±2.2	0.90±0.46	0.6±0.4	0.15±0.08	241±16	0.72±0.29	1.2±0.7	0.98±0.40
8/23/2002	24.3±3.1	0.47±0.40	0.2±0.2	0.05±0.04	293±25	0.39±0.24	0.3±0.3	0.41±0.26
6/18/2003	20.3±0.4	1.21±0.10	1.0±0.1	0.18±0.02	219±29	0.77±0.41	1.3±1.0	0.99±0.54
6/25/2003	27.0±1.3	1.53±0.37	1.6±0.6	0.33±0.08	297±26	0.87±0.46	1.7±1.3	1.63±0.87
Summer ave.	31.9±1.4	0.88±0.26	0.5±0.2	0.18±0.05	332±14	0.72±0.21	1.1±0.5	1.34±0.39
Sunset Beach								
6/5/2002	71.4±8.1	2.82±1.33	5.60±3.72	1.81±0.88	524±48	1.88±0.71	7.86±4.20	7.83±3.05
7/26/2002	55.3±5.2	0.83±0.49	0.49±0.40	0.25±0.15	727±39	0.60±0.20	0.80±0.38	1.86±0.63

Table 3
(A) Summary of samples collected during ebb tide near the mouth of each estuary (sample depth = 0.2 m), (B) Ra concentration of water following into each estuary, and (C) water budget and Ra flux from tidal exchange with each estuary (units = 10^6 atoms s^{-1} for ^{223}Ra and ^{224}Ra and 10^9 atoms s^{-1} for ^{228}Ra)

Station	Date and time collected	^{223}Ra	^{224}Ra	^{228}Ra
(A) Marsh Ra concentration (dpm m^{-3})				
Anaheim Bay	5/23/2003, 7:45	55±6	378±11	
San Gabriel River	5/23/2003, 8:30	36±4	296±9	62±4
Talbert Marsh	10/23/2000, 11:20[a]		550±80	111±29
Talbert Marsh	12/3/2000, 16:05			215±24
Talbert Marsh	3/28/2002, 10:00	58±3	506±7	
Talbert Marsh	5/23/2003, 7:00	46±6	333±12	
Santa Ana River	10/23/2000, 10:15[a]		460±50	93±21
Santa Ana River	12/3/2000, 15:20			184±20
	Average[b]:	49±5	421±41	133±29
	Dist. offshore to entrance (km)	^{223}Ra	^{224}Ra	^{228}Ra
(B) Ra input concentrations (dpm m^{-3})				
Alamitos Bay	0.90	37±4	247±25	60±6
San Gabriel River	0.30	47±5	394±39	76±8
Anaheim Bay	0.90	37±4	247±25	60±6
Talbert Marsh	0	32±3	338±34	79±8
Santa Ana River	0	32±3	338±34	79±8
Newport Bay	0.35	22±2	206±21	66±7
	Exchange rate (10^6 m^3 day^{-1})	^{223}Ra	^{224}Ra	^{228}Ra
(C) Water and Ra fluxes				
Alamitos Bay	3.41	12±12	52±31	12±11
San Gabriel River	0.52	0.3±1.9	1.2±5.0	1.5±1.7
Anaheim Bay	8.02	28±28	122±73	29±26
Talbert Marsh	0.52	2.5±1.8	3.8±4.9	1.4±1.7
Santa Ana River	0.87	4.1±3.0	6.3±8.1	2.3±2.8
Newport bay	10.1	77±34	191±91	34±33

[a]For ^{224}Ra, samples were processed and counted by gamma spectrometry following the method of Elsinger et al. (1982).
[b]Uncertainty computed as standard deviation of the mean.

explained by variations in lateral advection, in either the longshore or cross-shelf directions.

5. 2D mixing model

5.1. Local setting

The nearshore Ra distribution is influenced by the Ra sources and cross-shelf mixing. To explore these relationships, a 2D finite-difference model was developed to compute the Ra isotope distribution in surface waters of San Pedro Bay. To begin, a schematic of mean circulation in San Pedro Bay based on previous research is shown in Fig. 7. During the summer, mean flow through San Pedro Channel is equatorward (Hickey, 1993). Assuming geostrophic flow, this is also evident in the increase of about 2 °C observed between the NOAA Los Angeles Harbor Buoy and the Scripps San Pedro Buoy (Colbert, 2004). As the current moves south, its eastern boundary, Palos Verdes, disappears, and flow spreads out across San Pedro Bay (Hickey, 1992). At Huntington Beach, the average longshore advection rate (u) ranges between 3 and 10 cm s^{-1} (Hickey, 1993; KOMEX et al., 2003; MBC, 2002; Noble et al., 2003). The most complete and recent measurement of summer currents at the south end of San Pedro Bay found an average nearshore advection rate of 5 cm s^{-1} that was parallel to shore, flowing towards the southeast (Noble et al., 2003).

Two components of this description are incorporated into the mixing model. First, longshore advection was towards the southeast. Second, the observed longshore flow parallel to the shoreline was extrapolated from southeast of Sunset Beach to Newport Beach (Fig. 7). Based on current speeds measured near Huntington Beach, Ra isotope budgets require that only a modest fraction of the water offshore from Sunset Beach can be entrained in the persistent coastal current. The balance of the water must originate offshore and have background Ra concentrations.

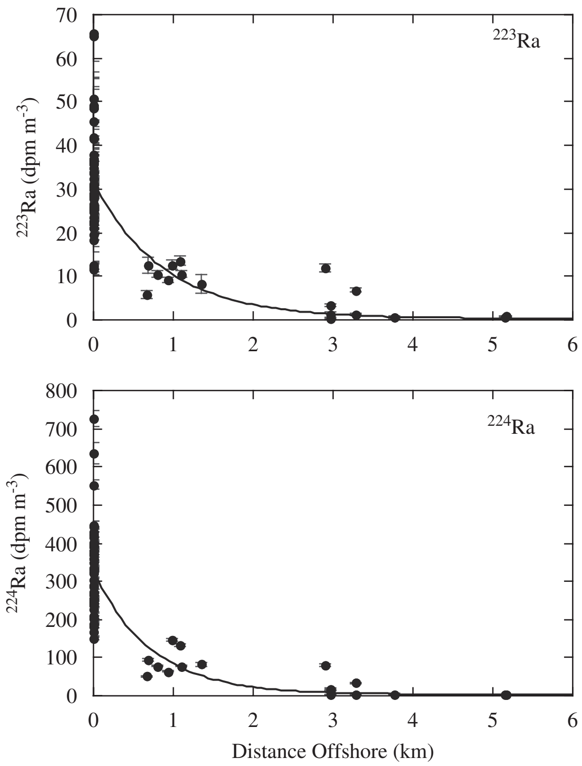

Fig. 6. Composite profile of summer data collected at Huntington Beach. Exponential fit (Eq. (2)) through data is shown. Model fit parameters are presented in Table 2.

5.2. Model geometry

The model covers 15 km of shoreline, from southeast of Sunset Beach to Newport Beach, extends 20 km offshore, and contains 30×80 cells (Fig. 1; Table 6). Longshore cells begin at the north end of the model and are all 500 m wide. Cross-shelf boxes begin at the shoreline, are 50 m at the shoreline and increase progressively by 5 m. The Huntington Beach transect is 9.5 km south of the upstream boundary.

Bathymetry within this region is fairly constant. The change in bathymetry between the shoreline and the mixed layer–seafloor intercept was incorporated into the model. The seafloor bathymetry was based on measurements made while sampling and NOAA Nautical Chart 18746 (Colbert, 2004). The mixed layer–seafloor intercept was 1.2 km offshore, defined by an average mixed-layer thickness of 11.8 m.

5.3. Boundaries

The model has three open boundaries. At the north and south boundary, a 1D, steady-state

Table 4
Sources of radium at Huntington Beach

	^{223}Ra	^{224}Ra	^{224}Ra/^{223}Ra flux ratio	Fraction of inputs	
				^{223}Ra	^{224}Ra
Shoreline flux					
Wave flux (1000 atoms s^{-1} (m shore)$^{-1}$)	0.06 ± 0.02	0.7 ± 0.2	12.2	1%	2%
Tide flux (1000 atoms s^{-1} (m shore)$^{-1}$)	0.36 ± 0.11	1.9 ± 0.5	5.2	5%	5%
Seafloor flux					
Ra flux (1000 atoms s^{-1} (m shore)$^{-1}$)	7.2 ± 3.6	39 ± 20	5.4	90%	93%
Tidal marshes and rivers flux					
Talbert Marsh (10^{6} atoms s^{-1})	2.5 ± 1.8	4 ± 5			
Santa Ana River (10^{6} atoms s^{-1})	4.1 ± 3.0	6 ± 8			
Length of coastline (km)	19	19			
Ra flux (1000 atoms s^{-1} (m shore)$^{-1}$)	0.34 ± 0.18	0.53 ± 0.50	1.5	4%	1%
Total input fluxes (1000 atoms s^{-1} (m shore)$^{-1}$)	8.0 ± 3.6	42 ± 20	5.3	100%	100%

Shoreline and seafloor fluxes are from Colbert and Hammond (2007).

Table 5
Estimate of inputs to the region south of San Pedro Bay

	^{223}Ra	^{224}Ra	^{224}Ra/^{223}Ra flux ratio	Fraction of inputs	
				^{223}Ra	^{224}Ra
Shoreline flux					
Wave flux (1000 atoms s^{-1} (m shore)$^{-1}$)	0.03±0.01	0.33±0.04	12.2	0%	2%
Tide flux (1000 atoms s^{-1} (m shore)$^{-1}$)	0.16±0.05	0.85±0.26	5.2	2%	4%
Seafloor flux					
Ra flux (1000 atoms s^{-1} (m shore)$^{-1}$)	0.76±0.11	4.13±0.59	5.4	10%	19%
Tidal marshes and rivers flux					
Newport Bay					
Ra flux (10^6 atoms s^{-1})	77±34	191±91			
Length of shoreline (km)	12	12			
Ra flux (1000 atoms s^{-1} (m shore)$^{-1}$)	6.4±3	15.9±6	2.5	87%	75%
Total input fluxes (1000 atoms s^{-1} (m shore)$^{-1}$)	7.4±2.5	21±6	2.9	100%	100%

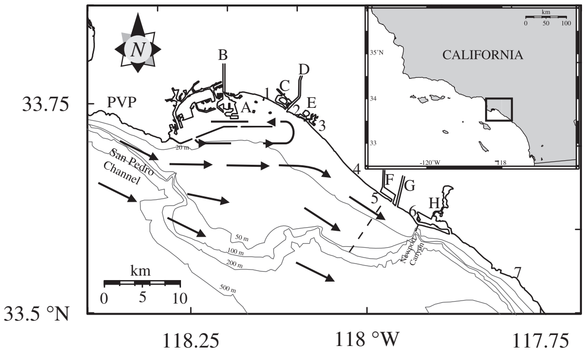

Fig. 7. Average summer circulation in San Pedro Bay, CA, with solid arrows based on previous work. Dashed line arrows are inferred from Ra balance based on this study. Arrows only represent direction of mean current, not current speed. Dotted line offshore from Huntington Beach is the location of current meter measurements made by Noble et al. (2003). Locations denoted by letters and numbers are the same as in Fig. 1.

Table 6
Model parameters used in the two-dimensional finite-difference model

Geometry	
Longshore boxes (*i*)	30
Offshore boxes (*j*)	50
Box longshore width (m)	500
Shoreline box offshore length (m)	50
Increase in box length offshore (m)	5
Seafloor gradient to 890 m offshore	0.01
Seafloor gradient beyond 890 m	0.00417
Mean mixed layer thickness (m)	11.8
Diffusive time step (s)	162
Fluxes	
^{223}Ra shoreline flux (1000 atoms s^{-1} (m shore)$^{-1}$)	0.42
^{223}Ra seafloor flux (1000 atoms s^{-1} (m shore)$^{-1}$)	7.2
^{224}Ra shoreline flux (1000 atoms s^{-1} (m shore)$^{-1}$)	2.6
^{224}Ra seafloor flux (1000 atoms s^{-1} (m shore)$^{-1}$)	39

Fluxes for San Pedro Bay are taken from Table 4.

model is solved that incorporates the seafloor geometry and assumes longshore changes in inputs and mixing rate are negligible. For the boundary model, the same K_h used in the model is applied. The role of boundary conditions on the Ra distribution within the model domain is an important component of this analysis and is further discussed below. At the offshore boundary, nominal ^{223}Ra and ^{224}Ra concentrations of 10^{-4} and 10^{-3} dpm m^{-3}, respectively, were defined.

At the base of the mixed layer, vertical transport of Ra by mixing or particles was ignored. Vertical mixing is inhibited by a strong density gradient. Based on the buoyancy gradient (Gargett, 1984), the loss by vertical mixing is at least an order of magnitude smaller than radioactive decay for the short-lived isotopes (Colbert, 2004). The short-lived Ra distribution further limits the vertical mixing flux. At Huntington Beach, more than 50% of the short-lived Ra is found between the shoreline and the mixed layer–seafloor intercept. Particulate removal of Ra would generate a vertical Ra gradient, particularly for ^{226}Ra. However, there is no ^{226}Ra activity gradient, horizontally or vertically, in this region (Colbert, 2004). Therefore, the Ra removal rate by particulates is less than the vertical mixing rate and must be negligible for the short-lived Ra budget.

5.4. Advection

Longshore currents are both a source and sink of Ra. For this study, three different longshore advection rates (u) were examined: 2, 5, and 7 cm s^{-1}. With 9.5 km between the model boundary and the transect used in the model (L), the transit time ($t_{\mathrm{Trans}} = L/u$) associated with each of these estimates was 5.5, 2.2, and 1.6 days, respectively. Alternatively, the residence time may be controlled by the longshore distance, L. Shorter residence times could be generated by a smaller L, which may occur if onshore advection occurs further south than Sunset Beach.

Mathematically, longshore advection is accomplished by moving the entire contents of a box in the desired direction of transport at a given time step, defined as the box width divided by the advection rate. To accommodate this, all boxes in the longshore direction have the same volume. Since the time step for advection is greater than the diffusion time step (hours vs. minutes), the advection time step is defined as an integer multiple of the diffusion time step.

Superimposed on the mean flow are tidal fluctuations that produce a northward flow during the flood tide and a southward flow during the ebb tide (Noble et al., 2003). For continuous sources, such as the shoreline and seafloor, tidal fluctuations have little impact since adjacent boxes have similar fluxes and concentrations. Therefore, the impact of tidal circulation on the Ra distribution can be ignored. However, tidal variations of the shoreline Ra input flux are important and are explored with this model (see Section 5.7.2).

5.5. Mass balance

For each cell, the change in mass of each isotope (ΔN) is solved for each time step (Δt):

$$\Delta N_{i,j} = (J_{\mathrm{dif}} + J_{\mathrm{in}} - J_{\mathrm{decay}})\Delta t = \Delta C_{i,j}\, V_j. \tag{4}$$

These include a diffusive exchange with up to four adjacent boxes (J_{dif}), an input flux (J_{in}) from the shoreline and seafloor, and a loss by radioactive decay (J_{decay}). $\Delta N_{i,j}$ is also equivalent to the product of the change in the box concentration ($\Delta C_{i,j}$) and the box volume (V_j). The subscript i defines position in the longshore direction and the subscript j defines cross-shelf position. The model is designed to find the distribution of isotopes at steady state, given a defined set of boundary conditions. For an advection rate greater than 1.5 cm s^{-1}, equilibrium is reached in less than 12 days. The model was verified for simplified cases by matching the model output to analytical solutions (Colbert, 2004).

5.5.1. Eddy diffusion

Diffusive exchange, J_{dif}, occurs between adjacent boxes:

$$J_{\mathrm{dif}} = -\sum A\, K_{\mathrm{h}} \frac{\Delta C}{\Delta x}, \tag{5}$$

where A is the cross-sectional area for the boundary between two adjacent boxes (m^2), K_{h} the eddy diffusivity (m^2 s^{-1}), ΔC the difference in concentration of two adjacent boxes (atoms m^{-3}), and Δx the distance between box midpoints. The summation represents the sum for all neighbor boxes (three for shoreline boxes and four elsewhere). K_{h} is the focus of this study. The diffusive exchange flux depends not only on the rate of eddy diffusion, but also on the cross-sectional area, which is a function of water depth. Thus, as the water depth increases offshore, the diffusive exchange rate, the product of the eddy diffusivity and the cross-sectional area, increases.

5.5.2. Radioactive decay

Radioactive decay is ubiquitous for Ra and is defined as

$$J_{\mathrm{decay}} = V_j\, \lambda_{\mathrm{Ra}}\, C_{ij}, \tag{6}$$

where λ_{Ra} is the decay rate for a given Ra isotope and V_{j} is the box volume.

5.5.3. Input flux

There are three sources of short-lived Ra to the mixed layer: estuaries, the shoreline, and the seafloor. Source functions (J_{in}) for each isotope have been measured (Table 4). Each box located between the shoreline and the mixed layer–seafloor intercept has an input from the seafloor, and the first row of boxes ($j = 1$) has an additional input from the shoreline. Inputs per unit area of seafloor decreased with distance offshore. Estuaries at the north end of San Pedro Bay are beyond the model boundary, and their importance is incorporated into the Ra distribution at the northern boundary. The Talbert Marsh and Santa Ana River have a negligible impact on the predicted Ra distribution at Huntington Beach because they are located downstream.

5.6. Data comparison

To compare the model results to the data, a routine was developed to compute the model-derived concentration at a given distance offshore (C_m) by assuming a linear concentration gradient between two adjacent box midpoints. The best fit to the data was found by minimizing the χ^2 statistic:

$$\chi^2 = \sum \left[\frac{C_{obs} - C_m}{\sigma_{obs}}\right]^2. \tag{7}$$

To simplify the calculation for the composite summertime Ra profiles, the observed concentration (C_{obs}) and its uncertainty (σ_{obs}) were averages at six points between the shoreline and 20 km offshore where sampling was concentrated. The mixing model was fit to those points for each isotope. Solutions for ^{223}Ra and ^{224}Ra were computed simultaneously, which allowed the χ^2 for each isotope to be combined to find the model parameters that best fit both data sets.

5.7. Model results and sensitivity

The mixing model was fit to the average summertime ^{223}Ra and ^{224}Ra profile at Huntington Beach for three advection rates, 2, 5, and 7 cm s^{-1}, by independently adjusting the mixing rate, K_h, and the inventory at the northern boundary, expressed as a fraction of the inventory at Sunset Beach, f_{SB}, until a best-fit combination was found (Fig. 8). For advection rates of 5 cm s^{-1}, the model computes an eddy diffusivity of 1.4 m^2 s^{-1}. The best-fit inventory at the upstream boundary of the model was equivalent to 8% of the inventory observed at Sunset Beach and 32% of the inputs at Huntington Beach, totaling 2.3×10^3 and 1.2×10^4 dps (m shoreline)$^{-1}$ for ^{223}Ra and ^{224}Ra, respectively. For slower model longshore advection rates, the best-fit eddy diffusivity increased and the inventory at the boundary decreased.

The uniqueness of each model solution was assessed by plotting a contour map of χ^2 for various combinations of K_h and f_{SB} (Fig. 9). The minimum value of χ^2 is located at the center of each plot, and the contours are integer multiples of the minimum. Assuming that $2\chi^2$ is equivalent to 1 standard deviation, then the uncertainty is 1.4 ± 0.4 m^2 s^{-1} for K_h and 8 ± 4% (absolute) for f_{SB}. Beyond $2\chi^2$, the contours are closer together, implying that the quality of the fit degrades quickly. This is shown

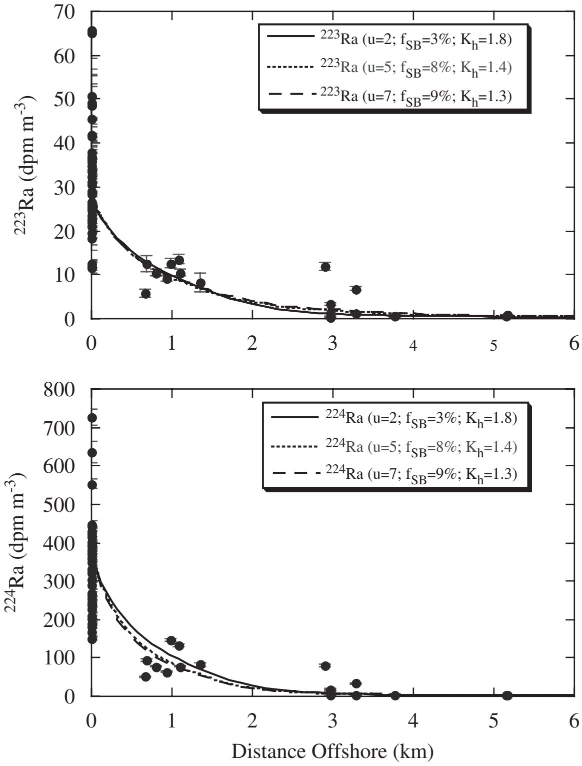

Fig. 8. Model sensitivity to mean longshore current velocity (u; cm s^{-1}). Composite summertime Ra profiles at Huntington Beach with model best-fit results for three current velocities. Model fit parameters are diffusivity (K_h; m^2 s^{-1}) and fraction of entrained Sunset Beach water (f_{SB}).

when f_{SB} is changed beyond 2 standard deviations of the average (Fig. 10).

For a constant f_{SB}, model sensitivity to eddy diffusivity was assessed. Results within the first 1.5 km offshore were most sensitive to changes in K_h (Fig. 11). Reducing K_h resulted in higher concentrations at the shoreline, while higher K_h reduced the concentration. Beyond about 2 km, changes in K_h had only a small impact on the concentration, with slightly higher concentrations as K_h increased. The region between 1.3 and 2.0 km offshore for ^{223}Ra and 1.0 and 1.5 km offshore for ^{224}Ra is the least sensitive to changes in K_h, with little variability in concentration over the range of K_h modeled. For a given transit time, to change the concentration significantly at about 1.5 km offshore requires a change in the Ra inventory at the upstream boundary (Fig. 10).

5.7.1. Model sensitivity

Model sensitivity was further assessed by changing several poorly constrained parameters within

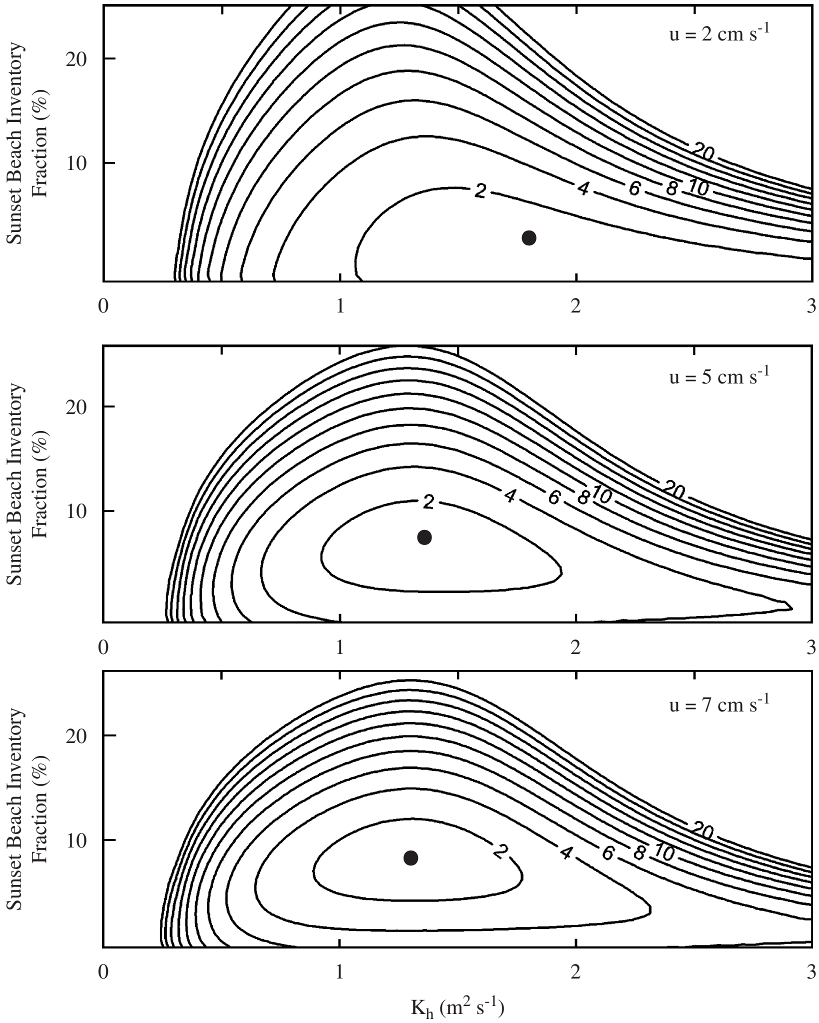

Fig. 9. χ^2 contour plots at three different advection rates as a function of mixing rate (K_h) and fraction of entrained Sunset Beach water (f_{SB}). Contours are spaced at integer multiples of the χ^2 minimum (approximately 7.5).

their uncertainty and computing the best fit to the average short-lived Ra profiles. First, the shoreline and seafloor inputs were reduced by 50%. At $5\,\mathrm{cm\,s^{-1}}$ the model accounts for the lost inputs by increasing f_{SB} to 14% to maintain the desired Ra inventory at Huntington Beach and decreasing K_h to $0.8\,\mathrm{m^2\,s^{-1}}$ to maintain high nearshore concentrations (Fig. 12).

Qualitatively, advection through the seafloor decreases as water depth increases (Huettel and Webster, 2001; Riedl et al., 1972), but an exact relationship has not been formulated. For the initial model, seafloor inputs were arbitrarily divided equally among the cells in contact with the mixed layer. Because the seafloor area in contact with each cell increased as box dimension increased offshore, the resulting seafloor flux per unit area decreased by 60% between the shoreline and the mixed layer–seafloor intercept; 45% of the inputs occur within the first 455 m. If inputs are concentrated nearshore, with 45% of inputs within the first 230 m, f_{SB} decreases and K_h increases (Fig. 13). As the inputs are further concentrated at the shoreline, the model fit further deteriorates, particularly for ^{223}Ra. To fit the observed profiles requires Ra inputs across seafloor in contact with the mixed layer.

The Ra distribution at the upstream boundary may be different than at Huntington Beach. For example, higher concentrations are observed further offshore at Sunset Beach. Or, an onshore flow may

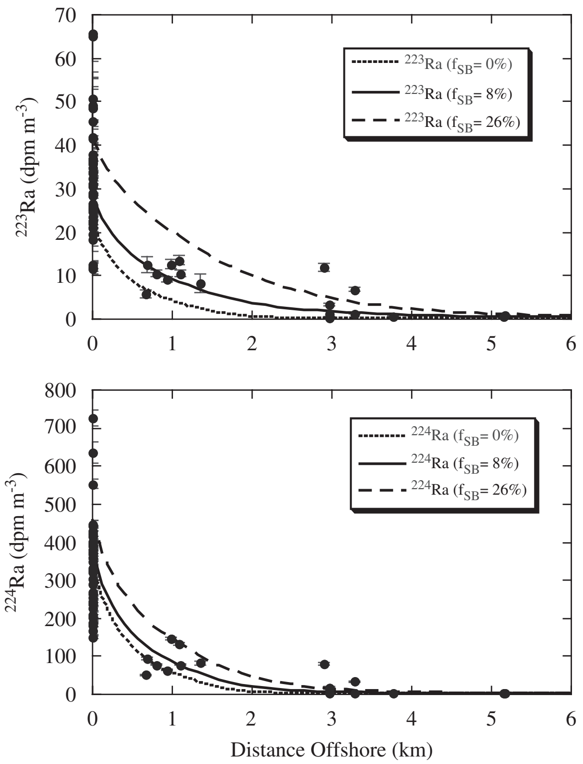

Fig. 10. Model sensitivity fraction of entrained Sunset Beach water (f_{SB}). For each case, advection rate is $5\,cm\,s^{-1}$ and K_h is $1.4\,m^2\,s^{-1}$.

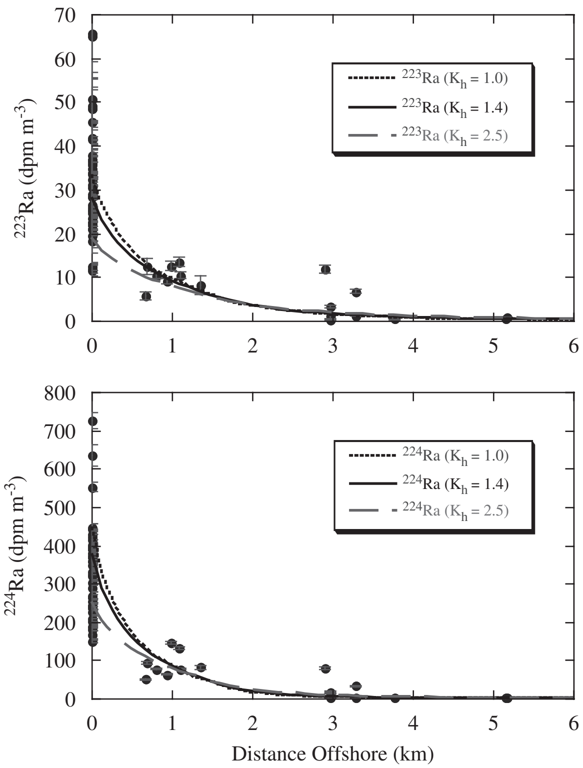

Fig. 11. Model sensitivity to eddy diffusivity (K_h; $m^2\,s^{-1}$). For each case, the advection rate is $5\,cm\,s^{-1}$ and the fraction of entrained Sunset Beach water is 8%.

bring low-Ra water into the nearshore. The offshore gradient at the upstream boundary was varied by changing the eddy diffusivity at the boundary while the inventory was held constant. For ^{224}Ra, the distribution at the upstream boundary had little impact on the distribution at Huntington Beach. ^{223}Ra was more sensitive, but the sensitivity decreases at longer transit times (Colbert, 2004).

5.7.2. Temporal variability

The nearshore mixing rate may vary on time scales of days to weeks, due to weather systems, lunar tidal cycles, and regional changes in circulation. On shorter time scales, diurnal winds and tides may also influence K_h. The time required to relax from one steady-state mixing regime to another was tested with this model. To minimize the impact of boundary conditions, the longshore box width was doubled to 1 km. At the shoreline, the short-lived Ra concentration responded quickly when K_h changed (Fig. 14). When K_h increases by a factor of 2.3, the shoreline concentration decreased by 10% after just 15 min. After 24 h, the shoreline ^{224}Ra concentration is about 80% of the way to the new equilibrium while ^{223}Ra is about 65%. At 1.3 km offshore, the change in concentration generated by the change in K_h is less than 5% (Fig. 15).

Tides impact the Ra flux from estuaries and water draining out of the beach. During the flood tide, the Ra flux should be minimal as water fills these reservoirs. During the ebb tide, water draining out of these reservoirs carries Ra into the coastal ocean. Discharge can even continue into the next flood tide, as inland water levels remain above sea level. To simulate this change in flux through time, the shoreline input flux was increased by a factor of 3, but only turned on for 1/3 of a tidal cycle (4.2 h). In Fig. 16, the resulting changes in concentration over 3 days (after a 30-day spin-up) is presented. Local changes in the shoreline input flux could account for about half of the observed fluctuations. In addition, these nearshore fluctuations are quickly averaged out, with identical concentrations predicted at less than 0.5 km offshore.

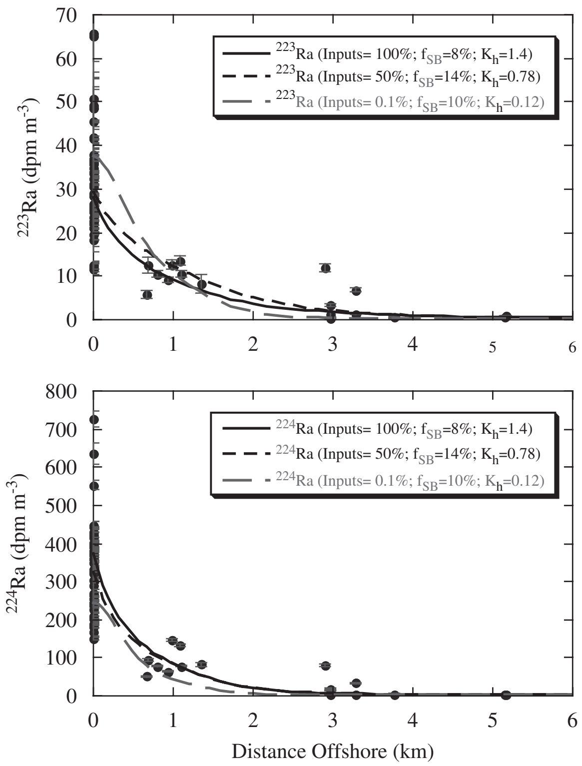

Fig. 12. Model sensitivity to seafloor inputs. Best fit to data given various input fluxes at a longshore advection rate of $5\,\mathrm{cm\,s^{-1}}$. Model fit parameters are diffusivity (K_h; $\mathrm{m^2\,s^{-1}}$) and fraction of entrained Sunset Beach water (f_{SB}).

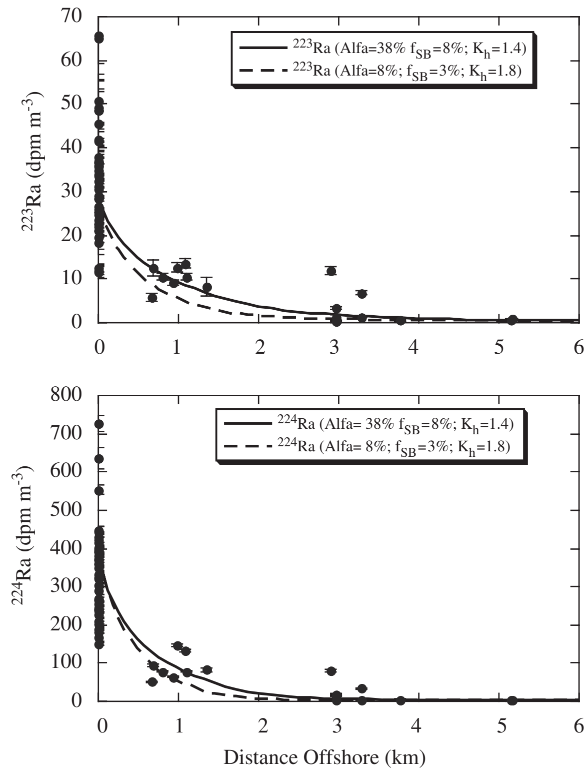

Fig. 13. Model sensitivity to location of seafloor inputs, varied with distance offshore. α is the ratio of seafloor input at furthest box, 1.2 km offshore, relative to the shoreline box. Best fit to data shown for a longshore advection rate of $5\,\mathrm{cm\,s^{-1}}$. Model fit parameters are diffusivity (K_h; $\mathrm{m^2\,s^{-1}}$) and fraction of entrained Sunset Beach water (f_{SB}).

6. Discussion

6.1. Temporal variability

The shoreline concentration varied at time scales of hours (Fig. 2). Two mechanisms that produce shoreline Ra concentration variability were identified. First, by varying the shoreline flux with the tidal phase, the model captures some of the observed shoreline concentration fluctuations. However, the model underestimates the total variability (calculated 15%, observed 25%), and does not account for higher frequency concentration changes. Second, the residence time of water in the surf zone, which is a function of K_h, may vary over short time periods. The surf zone is sensitive to K_h changes because of its relatively small volume and large Ra input flux. There is some evidence for a change in the nearshore mixing rate at various time scales. On a time scale of hours, onshore winds during the afternoon may reduce the rate of offshore transport (Noble et al., 2003). This is evident by the warming observed in the surf zone (Fig. 3). Surf zone circulation may fluctuate on shorter time scales. For a bar-trough morphology like the Huntington Beach surf zone, the surf zone water residence time fluctuates at the period of infra-gravity waves (0.5–4 min) and possibly at longer periods (MacMahan et al., 2004). Tidal changes in the trough volume may also change the residence time of water. However, it is unclear whether these fluctuations are sufficient to produce the large difference in mixing rate required to produce the observed concentration fluctuations.

Over periods greater than a week, the short-lived Ra inventory in surface water at Huntington Beach varied by up to 50%. Two mechanisms impact the Ra inventory. First, the inventory is sensitive to the longshore advection rate. Using the best-fit values for K_h and f_{SB} and changing the advection rate between 2 and $7\,\mathrm{cm\,s^{-1}}$, the inventory changes relative to the best-fit average by 18% and 22%

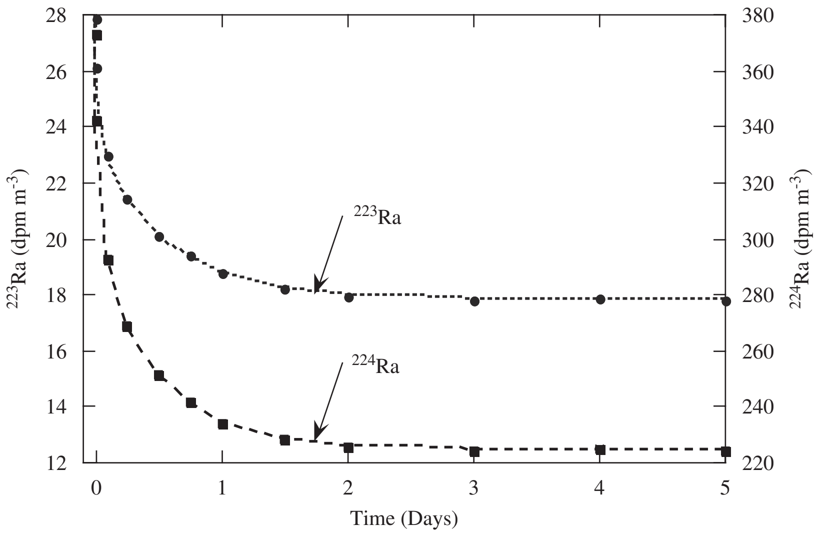

Fig. 14. Shoreline concentration as a function of time after a change in the mixing rate from $K_h = 1.4$ to $K_h = 3.0\,m^2\,s^{-1}$. Longshore advection rate is $5\,cm\,s^{-1}$ and f_{SB} is 8%.

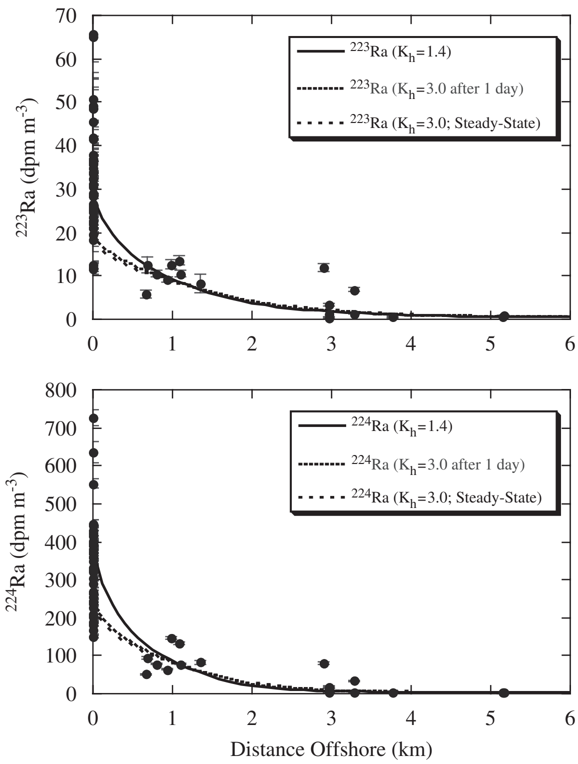

Fig. 15. Model response to a change in the eddy diffusivity (K_h). Model results are shown for steady-state $K_h = 1.4\,m^2\,s^{-1}$, profile 1 day after increasing K_h from 1.4 to $3.0\,m^2\,s^{-1}$, and steady-state $K_h = 3.0\,m^2\,s^{-1}$.

for ^{223}Ra and ^{224}Ra, respectively. To account for the larger inventory changes requires changing the fraction of water transported downcoast from Sunset Beach (for example Fig. 10). Based on this model, the range of inventories measured at Huntington Beach could be accounted for by varying the upstream boundary inventory between 0% and 25% of the inventory observed at Sunset Beach. For small changes in the inventory, it is not possible to distinguish between these two possibilities with the given data set. Regardless of the longshore advection rate, the fraction of water advected from Sunset Beach is less than 25% of the total flow.

6.2. Scale-dependent mixing

The K_h of $1.4 \pm 0.4\,m^2\,s^{-1}$ computed from the short-lived Ra data is consistent with other nearshore estimates. Mixing in the surf zone is controlled by breaking waves (Inman et al., 1971). Applying the formulation of Inman et al. (1971) to Huntington Beach, with a surf zone width (80 m), root-mean squared breaker height (0.5 m), and the spectral peak of the wave energy spectra (5 or 12 s, depending on the source), K_h is expected to range between 3 and $8\,m^2\,s^{-1}$. Diffusion in the surf zone decreases with depth (Inman et al., 1971), so this surface water estimate is an upper limit. Similar values have also been measured at the head of rip currents, where the change in diffusive processes from breaking waves in the surf zone to transport by eddies of various sizes most likely leads to a decrease in the diffusivity outside of the surf zone (Johnson and Pattiaratchi, 2004).

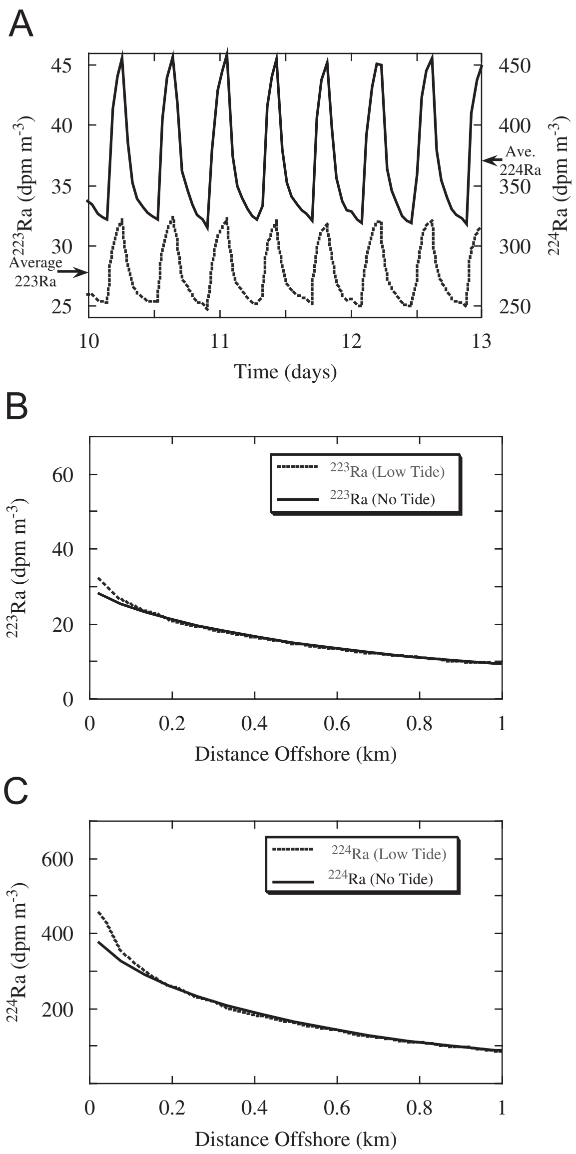

Fig. 16. Model response to tidally-driven changes in the shoreline flux ($K_h = 1.4\,m^2\,s^{-1}$). (A) Predicted shoreline variability at Huntington Beach over 3 days for ^{223}Ra (dashed line) and ^{224}Ra (solid line). Average concentration for each is shown on the *Y*-axis. The lower panels show the offshore distribution of (B) ^{223}Ra and (C) ^{224}Ra at the start of day 13 (dashed line) compared to the offshore Ra distribution predicted using when the shoreline flux is averaged over the entire tidal cycle.

However, the K_h estimate at Huntington Beach is orders of magnitude smaller than estimates using short-lived Ra tracers further offshore (Moore, 2000). The difference in these estimates may be explained if K_h increases with distance offshore. Since the nearshore is the primary source for Ra, this would be analogous to scale-dependent diffusion (Okubo, 1971).

The model was adjusted to account for an increase in mixing rate with distance offshore, by incorporating the following relationship:

$$K_h = 3.825x_m^{1.15} \text{ for } x \leqslant x_m, \tag{8a}$$

$$K_h = 3.825x^{1.15} \text{ for } x > x_m, \tag{8b}$$

where x is distance offshore in km, and the constants are as defined by the empirical observations of Okubo (1976). Eq. (8a) defines a constant K_h near the shoreline, and Eq. (8b) indicates scale-dependent mixing begins at some unknown distance offshore, with x_m left as an independent variable. By measuring distance in Eq. (8) from the shoreline, the minimum distance offshore to scale-dependent mixing is computed. The second variable was the inventory at the upstream boundary, f_{SB}.

A good fit to the short-lived Ra data was obtained using the model to reduce the χ^2 statistic (Fig. 17). The best fit to the data allowed for scale-dependent mixing beyond 455 ± 75 m, where the uncertainty is

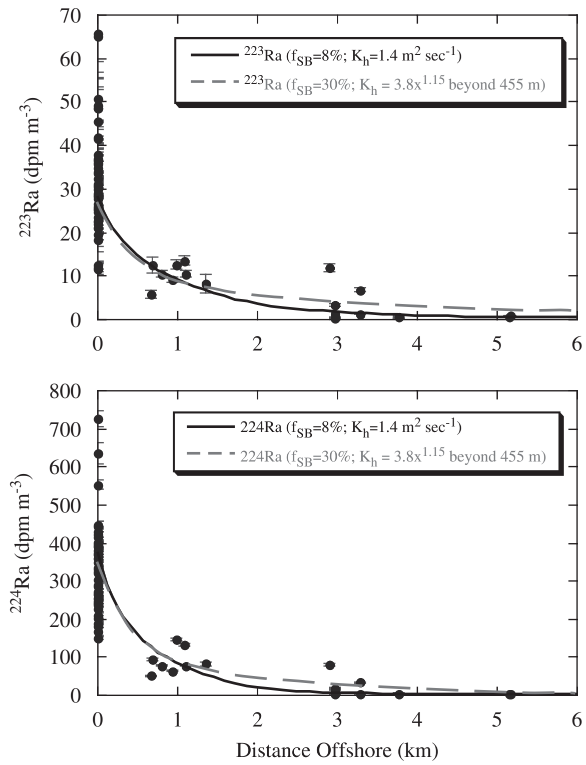

Fig. 17. Model generated profiles for constant-K_h (solid line) and scale-dependent K_h (dashed line). Longshore advection is $5\,cm\,s^{-1}$. Model fit parameters are diffusivity (K_h; $m^2\,s^{-1}$) and fraction of entrained Sunset Beach water (f_{SB}).

defined by the box spacing. Within the first 455 m, the mixing rate of $1.5\,m^2\,s^{-1}$ is similar to the constant-K_h estimate of $1.4 \pm 0.4\,m^2\,s^{-1}$. Beyond this, an increase in K_h produces a profile that is consistent with the observed data.

6.3. 1D vs. 2D results

The 1D model predicts an average K_h between 0.5 ± 0.2 and $1.1 \pm 0.5\,m^2\,s^{-1}$, based on ^{223}Ra and ^{224}Ra, respectively (Table 2). This isotope dependence of the mixing rate lends some doubt to its accuracy. For ^{223}Ra, onshore advection of low-Ra water and its longshore advection is an important part of the isotope budget, resulting in an underestimate of the mixing rate. The similarity of the ^{224}Ra result with the 2D model result of $1.4\,m^2\,s^{-1}$ and the variable-K_h model of $1.5\,m^2\,s^{-1}$ within the first 455 m is striking. Part of the success of the 1D model with the ^{224}Ra data is because the average inventory and input fluxes were similar. But, on individual sampling days, both the inventory and K_h varied, with higher inventories on days with higher K_h (Table 2). These different K_h estimates do not represent the temporal variability of the mixing rate. Instead, they are artifacts generated by changes in onshore flow of water into San Pedro Bay.

6.4. ^{228}Ra budget

Moore (1987) showed that without knowing the residence time of water on the shelf, the ^{228}Ra inventory underestimated its input flux because seasonal upwelling brought low-^{228}Ra water to the surface. For this study, the ^{228}Ra input flux can be estimated since the residence time of water, the offshore mixing rate, and the inventory of Ra in water entering the San Pedro Bay have been constrained. The eddy diffusivity and inventory as a function of the inventory at Sunset Beach were defined from the fit to the short-lived Ra data for both the constant-K_h and variable-K_h models. Water moving onshore was assumed to have a similar activity to that measured about 20 km offshore, $8\,dpm\,m^{-3}$. The seafloor and shoreline input fluxes can then be adjusted to find the best fit to the mean summertime ^{228}Ra surface water profile (Fig. 5). The model tends to underestimate concentrations at 1 km offshore, which suggests the presence of an additional source, such as upwelling. The best fit for the constant-K_h model required a seafloor input flux of $4.5 \times 10^6\,atoms\,s^{-1}$ (m shoreline)$^{-1}$. For the variable-K_h model, the best fit was found with an input flux of $3.4 \times 10^6\,atoms\,s^{-1}$ (m shoreline)$^{-1}$, with 12% originating at the shoreline and the remaining 88% from the seafloor. The volume of sands filtered to generate this input flux can be estimated. At Huntington Beach, the total sediment ^{228}Ra activity is $1.93\,dpm\,g^{-1}$, and the Ra emanation efficiency was found to be half-life dependent due to sorption within micropores (Colbert, 2004). Assuming the ^{228}Ra emanation efficiency is similar to ^{223}Ra (5.7%) and all of the ^{228}Ra is removed, a thickness of 90–130 cm of sand is required to generate the supply. Since ^{228}Ra is the first isotope in the ^{232}Th decay series, its emanation efficiency may be 50% greater than the short-lived isotopes (Krishnaswami et al., 1982). With this efficiency, the ^{228}Ra benthic input flux could be generated by removing all of the ^{228}Ra released from surface 60 to 90 cm of sand. This is considerably thicker than the estimate of 28 cm based on the pore water distribution of ^{222}Rn in these sands (Colbert and Hammond, 2007). However, the sensitivity of ^{222}Rn to measure circulation of seawater through sediments is limited to processes occurring on time scales of less than 20 days, while ^{228}Ra is sensitive to processes occurring at much longer time scales (months to years).

7. Conclusions

Profiles of ^{224}Ra and ^{223}Ra perpendicular to the coastline at Huntington Beach decreased offshore, and can be reasonably described by exponential functions. However, the eddy diffusivity calculated from each isotope are not concordant, due to complexities of the Ra source function. In addition, the inventories of each isotope increase with the calculated 1D eddy diffusivity, reflecting the importance of lateral advection in this setting.

In San Pedro Bay, the seafloor was identified as the primary source of short-lived Ra to the mixed layer, while estuaries at Huntington Beach and the shoreline were minor components. South of San Pedro Bay, the primary source of Ra to the mixed layer was the Newport Bay estuary. This was partly because of the large size of Newport Bay. However, the shoreline and seafloor sources were also reduced by low Ra production rates and a steeper seafloor slope.

A 2D diffusion–reaction–advection model was developed to compute the nearshore steady-state Ra distribution in surface water between Sunset Beach and Newport Beach, CA. The input fluxes were based on measurements made at the shoreline and seafloor offshore of Huntington Beach (Colbert and Hammond, 2007). To generate the best fit to the data required an input of Ra across the region in contact with the seafloor. Using a mean longshore advection rate of $5\,\mathrm{cm\,s^{-1}}$, the mean ^{223}Ra and ^{224}Ra surface-water offshore profiles could be explained with a horizontal eddy diffusivity rate of $1.4 \pm 0.4\,\mathrm{m^2\,s^{-1}}$, and an inventory at the northern boundary that is 8% of the observed surface water inventory offshore from Sunset Beach.

Over time scales of days to weeks, the short-lived Ra inventory in surface water at Huntington Beach was relatively constant. However, over longer periods, the inventory changed by as much as 50%. Changes in the short-lived Ra inventory at Huntington Beach may reflect changes in the longshore advection rate. As the advection rate increases, the nearshore residence time of water and the short-lived Ra inventory at Huntington Beach decrease. This can account for inventory changes of about 20%. To account for larger inventory changes at Huntington Beach requires a change in the fraction of water advected downcoast from Sunset Beach.

The shoreline short-lived Ra concentration varied by about 25% during tidal cycles, which could be explained with two mechanisms. First, when the shoreline input flux is concentrated during the ebb tide, the shoreline concentration fluctuated by 10–15%. Second, fluctuations in the offshore mixing rate may contribute to the shoreline concentration variability. Modest changes in the mixing rate can change the shoreline concentration by 20% in less than 1 h.

The measured mixing rate was similar to previously measured rates from the surf zone, where mixing is controlled by waves breaking, and the inner shelf. However, mixing in the open ocean is known to be scale dependent, increasing as a function of distance from the source of a tracer. With a nearshore source of Ra, scale-dependent mixing must begin at some distance offshore. After defining the increase in mixing from the shoreline in the model, scale-dependent mixing may occur beyond 455 m offshore. Between this point and the shoreline, the model-derived mixing rate of $1.5\,\mathrm{m^2\,s^{-1}}$ was similar to the constant-K_h model result.

To estimate the ^{228}Ra input flux to the summertime mixed layer, the constant-K_h and variable-K_h models were applied using the mixing parameters defined by the short-lived isotopes. Both of these models generated reasonable fits to the data. Depending on the emanation efficiency of ^{228}Ra, about 60 cm of sediment must be flushed to support the observed surface water inventory.

Acknowledgments

This project was funded by the University of Southern California Sea Grant Program, part of the National Sea Grant College Program, National Oceanic and Atmospheric Administration, US Department of Commerce, under Grants NA86RG0054 and NA16RG2256, and received additional support from the Wrigley Institute for Environmental Studies and the USC Center for Sustainable Cities. Gerry Smith, Carole Bartel, Altay Hammond, Timur Hammond, Erica Difilipo, and Shelley Howard helped with sample collection and analysis. Ra analysis expertise was provided by Ralph Arnold and Billy Moore. This project benefited from discussions with Shangde Luo, Richard Ku, Don Gorsline, Burt Jones, and Jed Fuhrman, as well as comments from two anonymous reviewers.

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